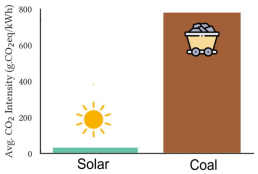
How Low Can We Go?: Analyzing Spatiotemporal Workload Shifting to Reduce U.S. Data Center Emissions

In support of Sukprasert et al., EuroSys 2024. **Mary-Alice Wieland**, Thanathorn Sukprasert, Walid A. Hanafy, Prashant Shenoy

Introduction

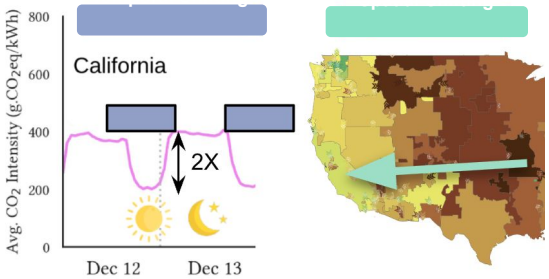
Carbon intensity (CI) represents the grams of carbon dioxide emitted per kilowatt-hour of electricity generated (gCO_2eq/kWh), serving as an indicator of how green a region's energy mix is. Combined with energy consumption, CI determines total **carbon emissions**.



Carbon Emissions (CO_2) = Energy consumed x CI

Background

Cloud applications are equipped with flexibility mechanisms that enables **workload shifting** to move tasks across regions and time.



Temporal Shifting reschedules tasks to times when CI is lower within a region.

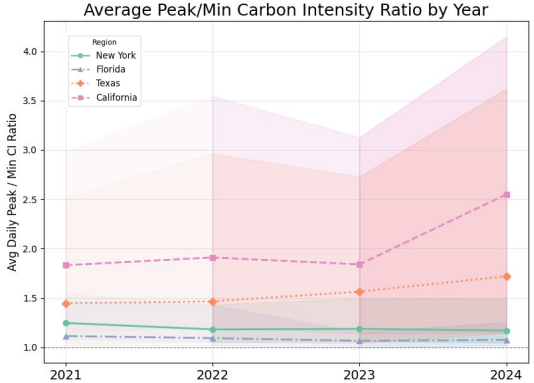
Spatial Shifting redirects tasks to other geographic regions with lower CI at a given time.

Methodology

We analyze carbon emissions from electricity use across U.S. cloud regions for spatial and temporal variability in carbon intensity (CI), measured in grams of CO_2 per kilowatt-hour. Our dataset is sourced from Electricity Maps, spanning the years 2021-2024.

Temporal Shifting

The intermittency of renewables causes CI to fluctuate over time. These fluctuations create opportunities for **temporal workload shifting** to reduce emissions.

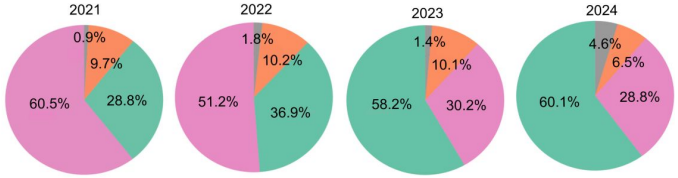


A higher ratio indicates greater temporal flexibility, or more opportunity to reduce emissions by running workloads during greener hours.

California and **Texas** show **high temporal variability** and would benefit from temporal migration.

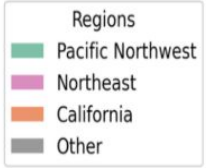
Florida and **New York** show **low temporal variability** and would benefit **less** from temporal migration.

Spatial Shifting

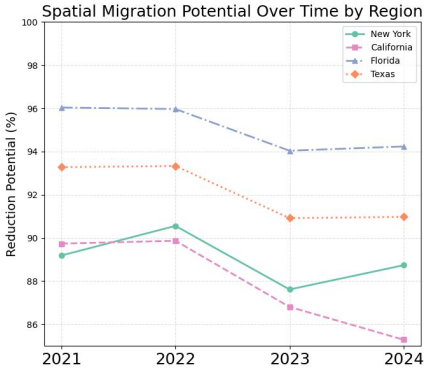


These regions maintain low CI making them the most reliable candidates for **spatial workload migration**.

"Other" regions grew from **0.9% in 2021 to 4.6% in 2024**, indicating more access to low-carbon electricity and a declining potential for reduction.



of



California: 90% to 85% reduction in CI from spatial migration.

Conclusion

As U.S. electricity grids decarbonize, regional differences in carbon intensity (CI) are narrowing, reducing the potential benefits of spatial workload migration. In contrast, temporal shifting may remain a more effective long-term strategy for curbing data center emissions, due to the high variability of renewable energy and the flexibility of cloud workloads.